Chapter 1 – (Week 2)

Section 1.3 Current in Semiconductors

1. Are free electrons in the valence band or in the conduction band?
2. Which electrons are responsible for electron current in silicon?
3. What is a hole?
4. At what energy level does hole current occur?

Section 1.4 N-Type and P-Type Semiconductors

1. Define doping.
2. What is the difference between a pentavalent atom and a trivalent atom?
3. What are other names for the pentavalent and trivalent atoms?
4. How is an n-type semiconductor formed?
5. How is a p-type semiconductor formed?
6. What is the majority carrier in an n-type semiconductor?
7. What is the majority carrier in a p-type semiconductor?
8. By what process are the majority carriers produced?
9. By what process are the minority carriers produced?
10. What is the difference between intrinsic and extrinsic semiconductors?

Section 1.5 PN Junction

1. What is a PN junction?
2. Explain diffusion.
3. Describe the depletion region.
4. Explain what the barrier potential is and how it is created.
5. What is the typical value of the barrier potential for a silicon diode?
6. What is the typical value of the barrier potential for a germanium diode?